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ALUMINATION: An app to connect with the Alumni's of the institute

¹ Likhit Yammanuru*, R. Harsha [†]
Department of Computer Science and Engineering
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham, India

*bl.en.u4cse22233@bl.students.amrita.edu, [†]bl.en.u4cse22249@bl.students.amrita.edu

Abstract—This project aims to create an alumni portal named Aluminatation, with distinct student and alumni interfaces, employing full-stack web development concepts to facilitate interactive discovery of alumni. The system has a Tinder-like swipe interface with ease-of-use navigation buttons, enabling students to browse alumni profiles according to hobbies, profession, and academic background. Through the use of scalable architecture with React.js, Node.js, and PostgreSQL, the portal is made modular, high-performance, and clean data-handling. Future integration with institutional ICTS systems is made possible by the portal, which also allows for secure OTP-based authentication and profile customization, thus promoting long-term academic and professional relationships.

Index Terms—Alumni dashboard, Alumni networking, Mentorship system, Event announcements, Real-time messaging, Institutional development.

I. INTRODUCTION

Current academic institutions do not have streamlined, interactive networks that enable students to link up with former students for mentorship, inspiration, and professional networking. With the growth of alumni communities, without the use of a centralized digital platform, students become disenfranchised from networks where they cannot easily learn about alumni paths, nor alumni contribute to student development. Existing alumni networks are outdated, cumbersome to use, or have limited features that do not promote dynamic interaction among students and alumni. A new, feature-filled alumni system is now required to close the gap between academic communities and their extended alumni network.

The research envisioned in this proposal involves a full-stack web application that constructs a Tinder-like interactive interface allowing students to browse alumni profiles and engage in meaningful interactions. Employing up-to-date web technologies like React.js for the frontend, Node.js for the backend, and PostgreSQL for relational data management, the system prioritizes modularity, performance, and intuitive design. The platform guarantees organized alumni discovery through swiping mechanisms paired with a neat and scalable design, providing customized browsing experiences via organized alumni data.

One of the largest challenges with alumni platforms is providing effective and engaging interfaces that scale with large datasets and support varied categories of users. Current

alumni portals are not typically responsive, personalized, or simply easy to navigate across huge sets of profiles. Current models are also not easily integrated with institutional databases or authentication mechanisms, further reducing their effectiveness in the academic community. Lacking scalable user interface flows and well-organized data pipelines, alumni sites end up being hard to maintain or add to for extended use.

This project suggests an alumni portal that makes a number of significant improvements to meet these challenges:

1. Implement a Tinder-like swipe interface through button-based navigation to easily discover alumni.
2. Student and alumni segregated interfaces to enable customized interactions and delivery of content.
3. Implement scalable backend and relational database systems to efficiently manage profile information and interaction records.
4. Support potential integration with institutional systems (e.g., ICTS) to automate account management and verification.
5. Provide speedy, interactive, and visually engaging UI experiences with high responsiveness and maintainability.

The project seeks to define a new standard for alumni-student engagement by developing an intuitive, scalable, and personalized portal to facilitate career exploration, mentorship, and long-term community building.

II. LITERATURE SURVEY

Young et al. (2024) [25], underscored the necessity of tailored tracking tools in milestone-focused graduate schooling. They noted that a milestones dashboard maximizes transparency, improves data consistency, and aids in informed decision-making throughout decentralized programs. Wei (2024) [24], investigated the application of HOG feature extraction in the analysis of student behavior in civic education. Through the integration of a learning dashboard and implementation of K-prototype and Kruskal-Wallis algorithms, the research determined the critical factors that promote motivation, engagement, and performance in blended learning settings.

Goss and Lui (2024) [9], investigated the effects of school climate on college and career readiness in California high schools. Employing a quantitative design, the authors established that caring relationships have a positive effect on readiness, with meaningful participation having a negative

¹Identify applicable funding agency here. If none, delete this.

correlation, with an emphasis on the importance of caring school communities. Carmichael et al. (2024) [16], probed the influence of personalized data dashboards on adherence in a 21-day EMA of drug use in college students. Though adherence was equal across groups, users finished surveys sooner and reported that messages were more compelling, implying heightened efficiency and engagement.

Bansak et al. (2024) [3], implemented student-initiated learning modules in macroeconomics to improve engagement through active research and analysis. Exercises such as policy-maker profiling, monetary policy debates, and FRED data analysis enhance critical thinking, data literacy, and understanding, even though they lack quantitative assessment. Afshari et al. (2023) [2], created a computer-based student dashboard for dental patient management using electronic health records. After implementation, the dashboard resulted in decreased overdue patients and increased compliance, with increased student engagement and quality of patient care.

Ismail et al. (2022) [11], developed an exploratory dashboard to show possible and existing alumni data in Universiti Teknologi MARA through multidimensional and hierarchical approaches. The dashboard was successful in showing employment and demographic information, although privacy limitations restricted the diversity of data, with suggestions for increased data fields in future studies. Hashish et al. (2025) [1], carried out a qualitative study to explore stakeholders' views on nursing education accreditation in a Saudi college. Six main themes were identified, showing the importance of accreditation in guaranteeing quality and employability, along with uncovering issues such as administrative load and the necessity of more robust institutional backing.

Cote et al. (2025) [6], surveyed alumni attitudes towards STEM experiences, and found concerns, misconceptions, and low confidence, with 86% unsure of their future in STEM despite previous exposure. Support systems such as STEM clubs promoted belonging and resilience, with an asset-based framework informing the research on community college students. Johnsen et al. (2024) [13], assessed the effect of critical thinking practice during dental school on application by graduates. Results indicated 60% utilized these skills on a daily basis, with 53% implementing them as individual procedures—most frequently in treatment planning—confirming the worth of incorporating critical thinking within dental education.

Jewell et al. (2025) [12], investigated the incorporation of practice change learning within the PharmD curriculum at The Ohio State University. Through interviews, focus groups, and curriculum mapping, the research identified areas for enhancement—including value-based outcomes and communication simulations—with a faculty lead tasked with spearheading curricular improvements. Holden et al. (2024) [10], reviewed King's College London's rapid-track dental program for medical graduates over a span of 14 years. Of 133 graduates, 70% followed careers of dual qualification—mostly OMFS (51.6%)—with low attrition and robust preparation calling for wider UK implementation.

Forrester et al. (2025) [8], examined Dickinson College's interdisciplinary method of developing a data analytics major, focusing on faculty initiative across disciplines, institutional backing, and careful curriculum development. The program, which had to operate under resource limitations, was able to successfully combine data science, ethics, and applied learning, with future growth planned and a proposed data science center. Dominguez and Ueno (2025) [7], recognized organizational impediments to campus support services for post-foster care youth, such as shortages of resources, bureaucratic barriers, and fragmented organizational structures. These interrelated problems impede program effectiveness, underscoring the necessity for enhanced institutional coordination and focused resource allocation.

Shania et al. (2024) [21], evaluated Kanban implementation in a software firm, citing challenges including project delays, WIP management, and low understanding. The research suggests targeted training and workflow improvements to mitigate these challenges. Rajput et al. (2024) [19], created a dashboard to support pre-medical students in tracking academic and extracurricular performance. Although faculty reaction was favorable, the study identified limitations such as lacking student feedback, formal usability testing, and implementation assessment.

McCulloch et al. (2021) [17], investigated the application of visualization methods to better comprehend issues for college students with autism. The research points to how these tools can make support systems more beneficial and facilitate increased accessibility in higher education. Pereira et al. (2022) [18], examined the way in which software startup experts make use of virtual Kanban boards for workflow management, enhancing communication, and monitoring progress. Although useful, the research established that most users struggled to use the tools in a useful manner.

Sanusi et al. (2023) [20], analyzed USM graduate conversion and employability using data analysis and machine learning. The study identified low PhD and MSc conversion rates with a focus needed on policy reforms to enhance academic progression and workforce participation. Khairunizan and Feisal (2023) [15], investigated the ways in which data quality enhancement enhances alumni tracking dashboards with emphasis on cleaning issues and new methods for record management efficiency and accuracy.

Kelly et al. (2023) [14], examined the influence of social and emotional support during college on foster care alumni post-college outcomes. Positive correlations with greater happiness and lower reliance on public aid were established, but there were no significant effects on work or income. Bettinger and Fidjeland (2024) [5], investigated the correlation between college rankings, alumni satisfaction, and labor market outcomes. The results indicated that students appreciate social experiences and personal development more than employment opportunities when considering their college experience.

Wang (2024) [23], investigated the ways to raise overseas alumni contributions through a coupled model. In their research, they discovered that tax incentives, networking, and

reputation drive giving, with tax benefits being the most significant driving force. Baumann and Halpern (2024) [4], examined the role of perceived value and satisfaction in influencing alumni loyalty. The research revealed that although alumni activities and institutional reputation increase perceived value, overall satisfaction is more instrumental in creating long-term loyalty.

Wang et al. (2022) [22], assessed a Learning Analytics Dashboard (LAD) embedded in the IIOE platform for online professional learning of college instructors. Participants liked and benefited from the LAD, although its effect on actual behavior change was minimal.

III. PROPOSED METHODOLOGY

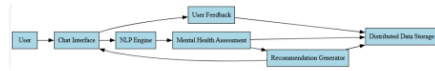


Fig. 1. System Overview of the Alumni Portal

The suggested system is a full-stack web application based on a modular architecture that aims to construct an interactive and scalable alumni portal that expands engagement between existing students and alumni. The system is based on a component-based architecture driven by React.js, Node.js, and PostgreSQL, providing dynamic discovery of alumni via a Tinder-like interface with the support of RESTful APIs and backend services.

A. System Architecture

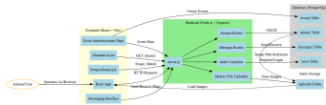


Fig. 2. Alumni Portal System Architecture

Portal architecture consists of three main layers combined to support a smooth user experience and future extensibility.

- **Frontend Layer:** Built with React.js and Vite, the frontend allows users to see alumni profiles in a swipeable interface using "Ahead" and "Change" buttons. Profiles show name, image, hobbies, job, and academic years.
- **Backend API Layer:** Developed on Express.js over Node.js, this layer supports RESTful interfaces for user login, alumni information retrieval, and upcoming features such as messaging and event announcement.
- **Database Layer:** Based on PostgreSQL to hold user credentials, alumni information, and related metadata. Security components such as bcrypt and JWT are implemented for safe login and session management.



Fig. 3. Tinder-style Swipe Interface for Alumni Discovery

B. Swipe Interaction Interface

- 1) **Swipe Simulation:** "Ahead" button simulates swiping forward through the list of alumni, while "Change" facilitates looping over choices.
- 2) **Mock Profiles:** Originally provided through hardcoded JSON in the server with placeholder data for prototyping use.
- 3) **Modular React Components:** Encapsulated components such as 'SwipeAlumni.jsx' contain rendering, state management, and event logic.

C. Two-Interface Model

The system is intended to deliver role-based experiences:

- **User Interface:** For students to browse and discover alumni with swipe logic and explore connections.
- **Alumni Interface:** To be built later, this will enable alumni to register, edit profiles, and communicate with students.

D. Authentication and Security

The login system has the following supported features:

- Password hashing using bcrypt.
- Token-based session management using JWT.
- OTP-based login to be integrated in the future.

E. Deployment and Testing

During the prototype stage:

- The frontend is run locally on Vite ('localhost:3000').
- Backend Node.js server hosts mock APIs ('localhost:5000').
- Postman is employed to test backend routes.
- Real alumni photos are to be served via a static hosting directory mapped to Express routes.

F. Evaluation Metrics

The system will be measured against the following:

- **Responsiveness** of the swipe interface.
- **Query Performance** from backend APIs.
- **User Satisfaction** through casual feedback and feature testing.
- **Maintainability** and simplicity of adding more features such as messaging and events.

IV. RESULTS AND DISCUSSION

The alumni-user interaction system was developed as a full-stack application with React for both the User and Alumni interfaces, sharing a common Express.js backend. The application provided mutual discovery, viewing of profiles, messaging, and announcements. All data (users, alumni, messages, announcements) was kept as mock arrays in the server code for demonstration.

a. User Interface Features: 1. Alumni Discovery: Users would be able to search alumni profiles through a swipe interface or search alphabetically by name through a specialized search page.

2. Profile Viewing: Alumni profiles had their name, city, interests, accomplishments, LinkedIn link, and profile image. Users would be able to see full details before sending messages.

3. Messaging System: Users would be able to send messages to any alumni. A floating animated message box with autofocus and toast confirmation was utilized to provide the best experience.

4. Inbox: Shows a list of alumni the user has messaged, with last messages and profile pictures retrieved dynamically.

b. Alumni Interface Features 1. User Discovery: Alumni might search users by name, with real-time filtering and clickable links to user profiles.

2. User Profiles: Profiles display name, registration number (in BL.EN.U4XXX2XXX format), semester, and profile image.

3. Messaging Users: Alumni might message users and see replies in a separate inbox, with proper user profile images and last messages.

4. Announcement Reception: Alumni were able to see a list of announcements (events, fests, etc.) received from the user interface in real time.

The outcomes of the system prove that the alumni-user interaction platform effectively facilitates two-way communication, profile discovery, and announcement sharing in a prototype setting. Users and alumni were able to send and receive messages, read profile information such as names and display pictures, and access announcements without problems. The Express backend shared for messaging routing and profile data across interfaces utilized in-memory arrays and local Storage for identifying senders. Although the UI provided a responsive and clean experience with swipeable cards and animated message modals, the lack of authentication and persistent storage restricts the scalability of the system for use beyond prototypes. Nevertheless, the implementation properly

proves the main concept and provides a solid base for further improvements.

V. CONCLUSION AND FUTURE SCOPE

Alumination gives students adynamic platform to engage with the alumni; engage in mentorship; network; and seek job opportunities on alumni circles. A trustworthy scalable easy platform powered by React.js, Node.js and PostgreSQL serves both students and alumni. The swiped-based system is structured to make simple connections between the students and alumni enabling to tighten bonds that may reach outside other college networks. When scaled appropriately and drawing in features, the platform is set for continuous growth. An upgrade oven is expected to include real-time communication, instantaneous notification of events, the use of OTP authentication, and the ability to maintain consistent interaction with institutional systems. With the use of these features, the platform will help increase the interaction of the users and strengthen Alumination's standing as a strong network of students at schools.

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